

Volume of Rectangular Prisms

Lesson 10-2

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

A box $2 \times 1.5 \times 1 = 3 \text{ ft}^3$ — it holds 3 cubic feet of stuff

Volume

How much space is inside a solid shape.

A tissue box, a fish tank, a brick — all rectangular prisms

Rectangular prism

A solid box shape with six flat rectangle sides.

1 ft^3 = a cube that is 1 foot on every edge

Cubic units

The units used to measure space inside, like cubic inches.

A box with dimensions $4 \times 3 \times 2$ means $l = 4$, $w = 3$, $h = 2$

Dimensions

How long, how wide, and how tall a shape is.

Cut a cereal box along its edges, unfold it flat — that is the net

Net

A flat shape that folds up into a solid.

If the base is $6 \times 4 = 24 \text{ cm}^2$ and $h = 3$, then $V = 24 \times 3 = 72 \text{ cm}^3$

Base area

The area of the bottom of a solid. Volume = base area $\times$ height.

Key Ideas & Notes

- Your class is building a time capsule to bury on school grounds.
- The capsule is a rectangular prism that is 2 feet long, 1.5 feet wide, and 1 foot tall.
- You need to figure out how much stuff can fit inside — that means calculating the volume!
- Calculate the volume of each possible time capsule design. Use $V = l \times w \times h$.

Think About It

- What are the three measurements of the time capsule?
- What does volume measure — length, area, or space inside?
- How is volume different from area?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Your time capsule is 2 ft long, 1.5 ft wide, and 1 ft tall — one edge is a fraction. Does having a fractional edge change how you find the volume? Why or why not?

Level 1 support

Start here: You still multiply length $\times$ width $\times$ height even when an edge is a fraction or decimal.

A fractional edge ____ **change the method because I still multiply** ____ $\times$ ____ $\times$ ____.

Una arista fraccionaria ____ *cambia el método porque sigo multiplicando* ____ $\times$ ____ $\times$ ____.

The volume of the capsule is ____ ft^3 .

El volumen de la cápsula es ____ ft^3 .

Word bank: volume dimensions rectangular prism base area cubic units

Listen for: Students explain the formula stays the same ($V = l \times w \times h = 2 \times 1.5 \times 1 = 3 \text{ ft}^3$) and that a fractional edge just means multiplying with a fraction or decimal.

Level 2

How could you use base area $\times$ height to find the same volume, and why does that give the same answer as $l \times w \times h$?

- Base area $\times$ height gives the same volume because base area is ____.
- First I find the base area ____ $\times$ ____ = ____, then multiply by the height ____.

EXPLORE

Capsule A ($2 \times 1.5 \times 1$) and Capsule B ($3 \times 1 \times 1$) both have a volume of 3 ft^3 . How did your multiplication produce the same volume from different dimensions?

Level 1 support

Start here: Different dimensions can multiply to the same product, so the volumes match.

Both capsules have $V = \underline{\hspace{1cm}} \text{ ft}^3$ because $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$.

Ambas cápsulas tienen $V = \underline{\hspace{1cm}} \text{ ft}^3$ porque $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$.

The dimensions are different but the $\underline{\hspace{1cm}}$ is the same.

Las dimensiones son diferentes pero el $\underline{\hspace{1cm}}$ es el mismo.

Word bank: volume dimensions rectangular prism base area cubic units

Listen for: Students show $2 \times 1.5 \times 1 = 3$ and $3 \times 1 \times 1 = 3$, and explain that the same product means equal volume even with different shapes.

Level 2

If two capsules have the same volume but different dimensions, which would you choose to store long, flat objects, and why?

- I would choose Capsule $\underline{\hspace{1cm}}$ because its dimensions $\underline{\hspace{1cm}}$.
- Same volume does not mean same usefulness because $\underline{\hspace{1cm}}$.

CONNECT

A fish tank is 24 in long, 12 in wide, and 16 in tall, and one gallon takes about 231 cubic inches. Talk through how you would find how many gallons it holds.

Level 1 support

Start here: Find the volume with $V = l \times w \times h$, then divide by 231 to convert to gallons.

The tank's volume is ____ **in³ because** $24 \times 12 \times 16 =$ ____.

El volumen del tanque es ____ *in³ porque* $24 \times 12 \times 16 =$ ____.

It holds about ____ **gallons because** ____ $\div 231 \approx$ ____.

Contiene unos ____ *galones porque* ____ $\div 231 \approx$ ____.

Word bank: (volume) (cubic units) (rectangular prism) (dimensions) (base area)

Listen for: Students compute 4608 in^3 , then divide by 231 to get about 20 gallons, and explain that dividing converts cubic inches into gallons.

Level 2

Why does the answer come out to about 20 gallons instead of an exact whole number, and how would you decide whether to round up or down for filling the tank?

- The answer is not exact because ____.
- To fill the tank safely I would round ____ because ____.

REFLECT

A classmate says a $5 \times 3 \times 2.5$ m prism has volume $5 \times 5.5 = 27.5 \text{ m}^3$. How do you find and explain their mistake?

Level 1 support

Start here: They added $3 + 2.5$ instead of multiplying; the correct volume is $5 \times 7.5 = 37.5 \text{ m}^3$.

The mistake is that they ____ instead of multiplying 3×2.5 .

El error es que ____ en lugar de multiplicar 3×2.5 .

The correct volume is ____ m^3 because $3 \times 2.5 =$ ____, then $5 \times$ ____ = ____.

El volumen correcto es ____ m^3 porque $3 \times 2.5 =$ ____, luego $5 \times$ ____ = ____.

Word bank: volume dimensions rectangular prism base area cubic units

Listen for: Students identify the added-instead-of-multiplied error, compute $3 \times 2.5 = 7.5$, and conclude the correct volume is 37.5 m^3 .

Level 2

What general rule about the volume formula does this mistake break, and how could a quick estimate have caught it?

- The mistake breaks the rule that volume always uses ____.
- An estimate would catch it because ____.

Guided Examples

Example 1

What is the volume of a rectangular prism with $l = 4$ in, $w = 3$ in, $h = 5$ in?

Solution: $V = l \times w \times h = 4 \times 3 \times 5 = 60$ cubic inches.

Answer: A. 60 in^3

Example 2

A box has a volume of 36 cm^3 . Its length is 6 cm and width is 3 cm. What is the height?

Solution: $V = l \times w \times h \rightarrow 36 = 6 \times 3 \times h \rightarrow 36 = 18h \rightarrow h = 2 \text{ cm}$.

Answer: A. 2 cm

Example 3

How is volume different from area?

Solution: Volume measures the space inside a 3D shape (cubic units like in^3). Area measures a flat 2D surface (square units like in^2).

Answer: A. Volume measures 3D space (cubic units); area measures 2D surface (square units)

Write About the Math

The Writing Revolution

I can explain my work using the words volume, rectangular prism, dimensions, and base area.

1. Kernel Sentence subject + verb

Model: Volume is how much space is inside a solid shape.

Volumen es cuánto espacio hay dentro de una figura sólida.

Write a kernel sentence about volume. Use a subject and a verb.

Escribe una oración base sobre volumen. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Volume matters in math

Volumen importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Volume matters in math because ____.

Volumen importa en matemáticas porque ____.

but
pero

Volume matters in math, but ____.

Volumen importa en matemáticas, pero ____.

so
entonces

Volume matters in math, so ____.

Volumen importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about volume.

Di un hecho verdadero sobre volume.

Volume ____.

Question *Pregunta*

Ask a question about volume.

Haz una pregunta sobre volume.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about volume.

Muestra entusiasmo sobre volume.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with volume.

Dile a un compañero qué hacer con volume.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

A fractional edge ____ **change the method because I still multiply** ____ × ____ × ____.

Una arista fraccionaria ____ *cambia el método porque sigo multiplicando* ____ × ____ × ____.

The volume of the capsule is ____ **ft³.**

El volumen de la cápsula es ____ *ft³.*

Both capsules have V = ____ **ft³ because** ____ × ____ × ____ = ____ × ____ × ____.

Ambas cápsulas tienen V = ____ *ft³ porque* ____ × ____ × ____ = ____ × ____ × ____.

Try It

Solve on your own. Check the answer key when you are done.

1. How is volume different from area?

- A. Volume measures 3D space (cubic units); area measures 2D surface (square units)
- B. Volume uses square units; area uses cubic units
- C. Volume only applies to cubes; area applies to all shapes
- D. There is no difference

Show your work:

2. Two boxes: Box A is $10 \times 5 \times 4$ inches. Box B is $8 \times 6 \times 5$ inches. Which has more volume and by how much?

- A. Box B by 40 in^3
- B. Box A by 40 in^3
- C. Box B by 60 in^3
- D. They're equal

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A toy company ships action figures in boxes that are $4 \times 3 \times 6$ inches. A shipping crate is $24 \times 12 \times 18$ inches. How many action figure boxes fit in the crate? Explain your reasoning step by step.

Sentence starter: Each toy box has a volume of ____ in^3 . The crate has a volume of ____ in^3 . I can fit ____ boxes because _____. I checked by _____.

Show your work:

Reflect — Exit Ticket

A rectangular prism has $l = 7$ ft, $w = 2$ ft, $h = 3$ ft. What is the volume?

- A. 42 ft^3
- B. 24 ft^3
- C. 42 ft^2
- D. 12 ft^3

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. Volume measures 3D space (cubic units); area measures 2D surface (square units) — *Volume measures the space inside a 3D shape (cubic units like in^3). Area measures a flat 2D surface (square units like in^2).*
2. **Try It 2:** A. Box B by 40 in^3 — *Box A: $10 \times 5 \times 4 = 200 \text{ in}^3$. Box B: $8 \times 6 \times 5 = 240 \text{ in}^3$. Box B is bigger by 40 in^3 .*
3. **Exit Ticket:** A. 42 ft^3 — *$V = 7 \times 2 \times 3 = 42$ cubic feet. Remember: volume uses cubic units (ft^3).*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Volume is how much space is inside a solid shape.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).